

Peter A. Lilley

Software Engineer - C#/.NET and C++17 Systems, Geometry and Automation Tooling

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SUMMARY

Software engineer. Writes C#, C++17, and Python against the Civil 3D, AutoCAD, and Revit APIs. Wrote C++ desktop applications and engine scripting before LLM tools existed. Constrains model output to a typed command schema, so deterministic code owns anything that has to be correct. Focus is applied ML systems and inference integration, not model training.

SKILLS

Languages: C#, Python, C++, Dart, TypeScript, JavaScript, SQL, LISP, VBA

Applied ML and inference: Structured model output with validation gates, Tesseract OCR, Levenshtein and fuzzy string matching, retrieval over structured corpora, COLMAP and instant-ngp photogrammetry, containerized inference, edge deployment, Ollama, LM Studio

Databases and Data Systems: SQLite, Firebase, Postgres, XData schemas, data modeling, JSON structuring, graph calculators

Engineering and math: Matrix math, 3D graphics, Express, React, Git, CMake, Linux, macOS, Docker, Bash scripting for data pipelines

Domain systems, quality, and networking: CAD plugins, Revit automation, 3D metadata, ARCore, part dimensioning, coordinate measuring machines, ISO 9001, TCP/IP, IPv4/IPv6 subnetting, Cisco routing and switching

EXPERIENCE

David Mason & Associates - Staff Technician (CAD automation / Civil 3D tools)

Jul 2024 - Present

St. Louis, MO

- Built a generator for standard details and grading that infers its output from a large body of existing company drawing data. Applied it to build the existing utilities model for an airport project, covering hundreds of thousands of surveyed entities. Logged project hours confirmed the work fell from months of hand modeling to days.
- Built a hydraulic grade line tool for Civil 3D, shipped as one DLL with a C++ core wrapped in C# and a WinForms interface. It draws pipe network water levels from an Excel source and edits pipe properties in bulk. Staff engineers documented the manual method at 4 to 8 hours per run and confirmed the tool does the same work in a few clicks.
- Built layout-wide edit commands that delete or move matching objects across every layout in a drawing, and freeze layers by drawing scale. A local phrase-to-command lookup picks the target layouts. Peers confirmed the manhour savings in a formal civil workload review.
- Write C# plugins against the Civil 3D .NET API to automate layout, audit drawings, and batch-rename title blocks. Typed metadata schemas attach to model entities and carry company standards across disciplines.
- Built OCR ingestion software that parses existing PDF plan sets, checks values, and populates drawings.
- Automate repetitive drafting with Python and Dynamo. In self-timed runs, one plan-production cycle dropped from 8 to 12 hours to about 30 seconds.
- Built the firm's internal Civil 3D automation pipeline on the OpenAI API. It serves a user chat, scheduled batch jobs, and a prepared store of company drawing data that conditions each request. The model returns command strings only. Deterministic C# code then constructs and validates the geometry and text, so a bad request fails validation instead of shipping a plausible wrong number. A local fallback serves the common commands when the API is unreachable.
- Extended that pipeline to locate outdated sheet text and to compare design options.

Component Bar Products - Quality Engineer - Manufacturing / Precision Machining

Apr 2023 - Jun 2024

O'Fallon, MO

- Programmed and operated coordinate measuring machines. Verified precision-machined parts against GD&T specifications.
- Validated dimensional tolerances against CAD models and prints to catch geometry mismatches.
- Enforced ISO 9001 across production lines. Managed calibration and traceability of metrology assets.

Menard, Inc. - Assistant Department Manager - Building Materials and Millwork

Jun 2020 - Mar 2023

Manchester / Ballwin, MO

- Priced and estimated large commercial material orders. Supervised department staff.

Heideman & Associates, Inc. - Revit/CAD Technician - Mechanical/Plumbing

May 2018 - Apr 2019

Fenton, MO

- Produced MEP construction documents. Built as-built models in Revit and AutoCAD.
- Built automated data libraries for building-system objects. Standardized outdated details.

PROJECTS

Digital Twin Pro

Personal product, Free Parameter LLC | Flutter/Dart 3, SQLite, Firebase, Google Gemini API, ARCore | github.com/freeParameterized/digital-twin-pro

- Designed and shipped a cross-platform 3D inventory application to a Google Play beta.
- Architected a custom 3D renderer with a Z-sorted draw queue and a custom projection instead of a game engine. Used LLM assistance to write most of the code.
- Added photo-based item detection through a paid Gemini API. Stored inventory in SQLite with a move audit log and CSV/JSON export.

CAD integration bridge

C++17, Dear ImGui, Windows COM | private repository, walkthrough on request

- Built a C++17 Dear ImGui host. It bridges BricsCAD, AutoCAD, and Civil 3D through Windows COM. It discovers installed SDKs at runtime and hot-loads LISP, C#, and BRX plugins.
- This personal project proved the architecture before a similar commercial deployment.

Interactive Portfolio (this site)

React, TypeScript, Express, Ollama, Docker

- Built a React and TypeScript front end over an Express API. The site builds to static files and needs no server to read.
- Containerized a local inference stack with Docker and ran it on Linux, macOS, and Windows hosts.

EDUCATION

Calculus II (CLEP Credit) - Saint Louis University (SLU)

2018

St. Louis, MO

Accepted by Ranken Technical College for their calculus curriculum.

Coursework toward Latin American Studies with Technical Applications - Missouri University of Science and Technology

2019 - 2020

Rolla, MO

Coursework included no-calculator College Algebra. It covered determinants, Cramer's rule, and introductory linear algebra concepts. Technical fluency in Spanish.

A.A.S., Building Systems Engineering Technology - Ranken Technical College

2019

Wentzville, MO